MIL-S-36113A
11 July 1972
SUPERSEDING
MIL-S-36113(MMSA)
14 November 1961

MILITARY SPECIFICATION

SIMMONS CITRATE AGAR, DEHYDRATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a dehydrated powder which, when prepared, is suitable for use as a medium for the differentiation of bacteria on the basis of citrate utilization. (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

PPP-C-00186a	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6505

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(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. Shall be a stable, dehydrated powder suitable for use as a medium for the differentiation of bacteria on the basis of citrate utilization. When prepared in accordance with the manufacturer's labeled directions, using purified water, shall contain in each 1000 cc of prepared medium, the following ingredients:

Magnesium Sulfate	0.2 Gram
Monoammonium Phosphate	1.0 Gram
Potassium Phosphate, Dibasic	1.0 Gram
Sodium Citrate	2.0 Grams
Sodium Chloride	5.0 Grams
Agar	15.0 Grams
Brom Thymol Blue	0.08 Gram

The medium shall contain not less than 95.0 percent and not more than 105.0 percent of the specified amount of each ingredient for each 1000 cc of prepared medium.

The pH of the prepared medium, after sterilization in an autoclave at 121° C. (15 pounds pressure) for 15 minutes, shall be between 6.70 and 6.90 at 25° C., when determined potentiometrically, using the U.S.P. procedure.

24.28 grams of dehydrated medium shall make 1 liter of Simmons Citrate Agar medium. One (1) pound (453.6 grams) shall yield not less than 18.5 and not more than 19.0 liters of Simmons Citrate Agar medium.

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3.2 Ingredients. All ingredients entering into the preparation shall be of the highest grade suitable for bacteriological use.

3.3 Contents. Filled immediate containers shall contain an average weight of not less than 1 lb (453.6 grams) and no one immediate container shall contain less than 431.0 grams, when tested as specified in 4.6.2.

3.4 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

3.5 Workmanship. Material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of Simmons Citrate Agar Dehydrated, produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final, immediate containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

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For visual examination	Inspection level*	AQL (percent defective)	Unit of product
Major A	S-2	1.0	Filled bottle
Major B	II	1.0	Filled bottle
Major C	II	1.0	Filled bottle
Major D	II	1.0	Intermediate package
Major E	II	1.0	Exterior package
Minor A	II	2.5	Filled bottle

4.3.1 For end item testing. Acceptance number shall be zero (0) for all sample sizes.

Characteristic	Requirement paragraph	Inspection level*	Minimum sample
pH (prepared medium)	3.1	S-2	5
Net contents	3.3	S-2	5
Differential capability	3.1	S-2	5
Yield	3.1	S-2	5
Label adhesion		per PPP-C-00186a	10

* Where possible, the same sample shall be used for two (2) or more tests.

4.4 Examination. The Simmons Citrate Agar, Dehydrated, shall be examined to determine compliance with all requirements of this specification.

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4.5 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Classification of Defects**

Major A	
101	Powder not free from caking.
102	Color of powder not homogeneous.
103	Powder not free of foreign matter.
Major B	
151	Labeling not complete.
152	Labeling not legible, not free from defacing marks and tears.
Major C	
161	Immediate container not as specified.
162	Immediate container not properly sealed.
Major D	
171	Material and marking of intermediate package not as specified.
Major E	
181	Material and marking of exteiror contianer not as specified.
Minor A	
201	Exterior of immediate container not free from contents or foreign material.

** Inspection is not restricted to the classified possible defects listed above.

4.6 Tests.

4.6.1 Tests and examinations shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.6.2 Weight of contents. Weight of contents shall be determined by a suitable method.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-0018ba, dated 15 May 1969, and Amendment-1, dated 27 October 1969, and as specified herein:

5.1.1 Immediate containers. Each immediate container shall comply with the following classification:

GROUP A	CLASS 1	TYPE e	STYLE 2	GRADE 1
CLOSURE A				SEAL A and B

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5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below:

5.2.1 Immediate containers. Each immediate container (packet) label shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) the item name designated as "SIMMONS CITRATE AGAR, DEHYDRATED"

(b) the quantity of contents designated as "1 lb (453.6 Grams)"

Note: The official abbreviation "g." may be used in lieu of the word "grams".

(c) the lot or control number

(d) the Federal Stock No. designated as "FSN 6505-153-8806" or "Stock No. 6505-153-8806"

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) directions for preparing the medium for use.

(g) the formula of the product for 1000 cc of prepared medium

(h) the date of manufacture.

(i) the statement "Store at controlled room temperature (59° - 86° F.)."

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5.3 Packaging and packing.

5.3.1 Unit of issue. One bottle containing 1 pound, as specified, constitutes one unit of issue.

5.3.2 Unit package. At the option of the contractor, each unit shall be packaged as specified in 5.2.5 of PPP-C-00186a.

5.3.3 Procedure code. Procedure code No. 6 of PPP-C-00186a applies.

5.4 Marking.

5.4.1 Unit package. When furnished, each unit package shall bear the same information as required for the label of the immediate container.

5.4.2 Intermediate package. In paragraph 5.5.3 of PPP-C-00186a, add: "Date of manufacture shall be shown in lieu of date packed. In addition, marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)."

5.4.3 Exterior container. In paragraph 5.5.4 of PPP-C-00186a, add: "Date of manufacture shall be shown in lieu of date packed. In addition, marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification
(b) the Federal Stock No.
(c) applicable levels of packaging and packing (see PPP-C-00186a).

6.2 Federal Stock No. coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification	Unit
6505-153-8806	SIMMONS CITRATE AGAR, DEHYDRATED, 1 lb (453.6 Grams)	Bottle

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1292

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-36113A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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Defense Supply Agency

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DOD-304

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ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
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PHILADELPHIA, PA. 19101

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